

PHYTOGUARD

REFINED PROJECT BLUEPRINT : R&D; : SAFETY : SCALABILITY : HACKATHON MANUAL

Open, affordable process intelligence for small-scale botanical R&D; - with a software control hub, hardware-adapter architecture, Wokwi simulation, experiment data layer and a low-cost reference controller.

Dimension	Integrated Project Position
Core product	A web-based control, monitoring, experiment-data and analytics hub.
Reference hardware	ESP32 + low-cost sensing/relay/display stack (Rs 1,120 BOM) proving the digital architecture.
Interoperability	Hardware adapter layer maps heterogeneous sensors/actuators into common software variables.
Simulation	Wokwi reference build allows users to test controller logic and wiring before physical assembly.
Safety philosophy	Local fail-safe on-chip control first (<10ms trip); cloud/web telemetry is secondary to physical interlocks.
Medical relevance	Enables safer, reproducible botanical R&D; does not claim to certify medicine or clinical efficacy.
Scale strategy	One platform -> many hardware profiles -> many experiments -> reusable research data.

Central Motto: "Turn affordable, heterogeneous electronics into a shared digital nervous system for small-scale botanical R&D;."

Prepared for: Team Zenith • EDGE NOVA'26 | **Version:** Refined Integrated Blueprint (September 2026)
Scope Boundary: This document is an engineering design and hackathon blueprint. It is not an industrial hazardous-area certification, clinical trial protocol, or medical device certification.

1. Reference Hardware Architecture & Rs 1,120 Prototype BOM

The physical controller acts as the open-source reference implementation that proves the larger software platform. It demonstrates how low-cost, off-the-shelf electronics can be safely interfaced with extraction apparatus.

Component	Reference Role	Qty	Cost (INR)	Safety & Interface Note
ESP32 38-Pin DevKit	Controller & Local State Engine	1	Rs 350	Dual-core LX6, 1-Wire, ADC1, I2C, Wi-Fi
5V Optocoupled Relay	Heater Mantle Switch	1	Rs 68	GPIO 26 (Active LOW). Normally-Open (NO)
MQ-3 Alcohol Sensor	Vapor Proxy / Hazard Signal	1	Rs 120	GPIO 34 (ADC1_CH6). Uncalibrated proxy
DHT22 (AM2302)	Chamber Temp & Relative Humidity	1	Rs 145	GPIO 4 (1-Wire digital with parity check)
0.96in I2C OLED (SSD1306)	Immediate Benchtop Display	1	Rs 158	I2C Bus (SDA 21, SCL 22). Local visual state
Active Buzzer + LEDs	Acoustic & Optical Alarms	1 set	Rs 19	GPIO 27 (Buzzer), GPIO 16 (Grn), GPIO 17 (Red)
2x Li-ion (1200mAh) + BMS	Isolated Logic Power Supply	1 set	Rs 211	Isolates MCU from mains inductive switching noise
Jumpers, Resistors, Board	Interconnects & Pull-ups	1 set	Rs 49	4.7k pull-up for 1-Wire bus conditioning
TOTAL REFERENCE BOM	Complete Prototype Hardware Stack	-	Rs 1,120	Low-cost accessible decentralized R&D;

Physical Design Safety Rule: Keep electronic switching hardware physically separated from solvent-vapor exposure. The Rs 1,120 BOM represents the prototype digital stack, not a certified explosion-proof industrial extraction suite.

2. System Logic - The 10 Integrated Architectural Layers

The system is structured into ten distinct architectural layers ensuring modularity, deterministic local safety, and high-level research data integrity:

Layer	Name	Core Function	Why It Exists
1	Physical Process	Extraction vessel, botanical sample, solvent, heating mantle, water condenser.	Real-world laboratory process observed and safeguarded.
2	Sensors & Actuators	DHT22, MQ-3 vapor proxy, optocoupled relay, active piezo buzzer, OLED.	Transforms physical environment into electrical signals and actions.
3	Local Controller	ESP32 microcontroller executing hard real-time loop at 1 Hz.	Guarantees <10ms local trip response independent of network status.
4	Hardware Adapter	Translates pin/ADC signals into standardized software variables.	Makes the web platform 100% hardware-agnostic.
5	Communication	Transmits JSON telemetry over Wi-Fi/MQTT/REST; handles connection dropouts.	Bridges physical node to the digital cloud hub.
6	Control Engine	Applies limits, safety interlocks, PID heater logic, and state transitions.	Enforces repeatable, documented process conditions.
7	Data Hub	Records synchronized timestamps, state flags, lot metadata, and events.	Creates traceable, audit-ready experiment records.
8	Analytics Layer	Overlays multi-run curves, computes thermal stability, grades reproducibility.	Transforms raw time-series into actionable scientific evidence.
9	User Interface	Responsive web dashboard, profile manager, live alert feeds, CSV/JSON export.	Enables intuitive control by botanical researchers and operators.
10	Simulation Layer	Wokwi reference virtual twin and fault-injection scenarios.	Reduces entry barriers and verifies code before bench assembly.

3. Hardware-Agnostic Architecture (Abstraction Layer)

A central innovation of PHYTOGUARD is decoupling the web dashboard and research analytics from component-specific firmware. Replacing a temperature sensor or switching relay never requires rewriting the platform.

The Abstraction Rule: The application layer only consumes normalized logical variables (*temperature_c*, *humidity_rh*, *vapor_signal*, *heater_state*, *alarm_state*, *fault_code*). The hardware adapter converts device-specific inputs (GPIOs, 1-Wire CRC, 12-bit ADC) into standard engineering units.

Profile Name	Controller	Temperature Sensor	Vapor/Gas Sensor	Actuator Switch
Profile A (Reference Stack)	ESP32 DevKit (38-Pin)	DHT22 (1-Wire GPIO 4)	MQ-3 (ADC1_CH6 GPIO 34)	5V Opto Relay (GPIO 26)
Profile B (Industrial Lab Pro)	ESP32-S3 Feather Dual-Core	DS18B20 RTD (GPIO 14)	Sensirion SGP30 (I2C 0x58)	SSR-40DA Opto Solid State
Profile C (Wokwi Virtual Twin)	Wokwi Simulated ESP32	Virtual DHT22 Model	Virtual Potentiometer ADC	Virtual Relay Indicator

4. Safety Architecture - Deterministic Local-First Interlocks

Safety cannot rely on remote web connectivity. Loss of Wi-Fi, cloud downtime, or browser latency must never stand between an overheating extraction vessel and a fail-safe shutdown.

Safety Hierarchy of Protection: *Physical Safeguards (Hood/Condenser) -> Local MCU Interlocks (<10ms) -> Local Alarms (Buzzer/OLED) -> Web Dashboard Telemetry -> Post-Run Analytics.*

4.1 The 5-State Machine Engine

State	Entry Conditions	Local Controller Action	Recovery Protocol
NORMAL	All critical sensors valid. Temp < 72.0 C, Vapor < 1800 ADC.	Normal closed-loop PID control; green LED on; stream telemetry.	Autonomous operating state.
WARNING	Temp >= 72.0 C or Vapor Proxy >= 1800 ADC.	Yellow LED on; log warning event; restrict heater duty cycle.	Auto-clears when readings return below threshold.
ALERT	Temp >= 78.5 C (Ethanol boiling hazard) or Vapor >= 2800 ADC.	DE-ENERGIZE RELAY IN <10MS ; trigger acoustic piezo alarm; red strobe.	Manual acknowledge + verified temperature < 65 C.
FAULT	DHT22 parity error, missing packets > 3.5s, or wire cut.	FAIL-OPEN RELAY TRIP ; mark data as UNCERTAIN; sound fault pattern.	Hardware diagnostic + sensor reconnect + reset.
LOCK	Operator E-STOP button or repeated safety fault.	Full hardware isolation; lock heater relay open; disable web overrides.	Physical inspection + manual key/button reset.

4.2 Critical Hazard & Flammable Liquid Standards (OSHA Alignment)

- Ethanol Vapor Ignition Risk:** Ethanol has a flash point around 55 deg F (13 deg C) and an explosive flammability range of 3.3% to 19% by volume in air.
- Non-Intrinsic Safety Disclaimer:** The Rs 1,120 reference prototype electronics and mechanical relays are not certified as intrinsically safe or explosion-proof. Extraction glassware, heat mantles, and electronics must be separated, with the process operating under active exhaust ventilation.

5. Sensor Strategy & R&D; Validation Workflow

Claim Discipline: Low-cost sensors such as the MQ-3 tin-oxide semiconductor respond to volatile organic vapors but must be documented as **uncalibrated hazard proxies** rather than certified concentration meters unless calibrated against an analytical gas chromatograph.

The 7-Step Sensor Calibration & Validation Pipeline:

- Baseline:** Record sensor resistance in clean ambient air.
- Reference:** Characterize response curve against calibrated laboratory reference instruments.
- Repeatability:** Execute multiple temperature cycles to verify signal hysteresis.
- Drift Evaluation:** Measure baseline resistance drift under prolonged heat/solvent exposure.
- Placement Study:** Optimize sensor distance above condenser outlet to avoid condensation wetting.
- Uncertainty Tagging:** Tag all telemetry records with calibration status (VERIFIED, PENDING, UNCERTAIN).
- Claim Discipline:** Transparently state sensor boundaries in published experiment reports.

6. Web Platform Blueprint - The Scalable Product

While the ESP32 handles local fail-safe execution, the web platform is the comprehensive software product that manages devices, runs experiments, analyzes trends, and enforces research reproducibility.

Module	Built-In Functionality	Research Gap Addressed
Live Dashboard	Live KPI cards, synchronized Recharts stream (1 Hz), threshold lines, manual overrides.	Eliminates fragmented, blind microcontroller telemetry.
Hardware Profiles	HAL mapping of GPIO pins and ADC channels to normalized variables.	Decouples software from specific microcontroller boards.
Safety Engine	Interactive 5-state machine visualizer, threshold limits manager, E-STOP latch.	Prevents dangerous thermal runaway and solvent leaks.
Experiment Manager	Run creator (botanical species, lot #, solvent, method), live recorder, notes.	Transforms sensor values into traceable scientific research records.
Research Analytics	Superimposed multi-run comparison, thermal stability index, reproducibility grading.	Provides rigorous evidence base for botanical development.
Wokwi Simulator Hub	Interactive wiring pinout, virtual twin testing, one-click fault injection sandbox.	Lowers onboarding barriers and prevents circuit damage.
Export & Audit Layer	RFC 4180 CSV export, structured JSON schemas, printable run dossier.	Ensures data portability into R, Python, and Excel.
Firmware Docs	Complete Arduino/C++ sketch viewer with copy button, REST & MQTT schemas.	100% open-source OSHWA compliant transparency.

Recommended 10-Step Researcher Workflow:

1. Create experiment record with botanical ID & solvent lot -> 2. Select hardware profile -> 3. Verify wiring in Wokwi simulator -> 4. Assemble bench hardware -> 5. Check sensor calibration status -> 6. Connect controller to web hub -> 7. Start run & monitor live stream -> 8. Local MCU enforces fail-safe limits -> 9. Review anomaly events & multi-run comparison -> 10. Export verified CSV/JSON research record.

7. Research Basis, WHO Alignment & Summary Positioning

- **WHO Global Traditional Medicine Strategy (2025-2034):** Directly addresses key pillars of research, safety, quality, evidence, innovation, sustainability, and equity.
- **Responsible Flora Framing:** Encourages documented, observable R&D; into indigenous botanical resources without falsely claiming medical certification or drug manufacturing status.
- **Closing Elevator Pitch:** "PHYTOGUARD turns affordable, heterogeneous electronics into a shared digital nervous system for small-scale botanical R&D; - simulate it, assemble it, monitor it, and learn from it."